

Effects of Robotic Prosthetic Hand Use on Muscle Fatigue and Motor Performance in Daily Activities

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Abstract—Muscle Fatigue (MF) is a significant challenge in upper-limb prosthesis, driven largely by device weight and prolonged wear; however, the full extent of fatigue-related consequences remains insufficiently explored. To the best of the authors’ knowledge, this study is the first to investigate MF during simulated Activities of Daily Life (ADLs) in able-bodied participants using an upper-limb prosthesis. Muscle activity, joint biomechanics, movement velocity, and Rated Perceived Exertion (RPE) were assessed to evaluate the impact of prosthesis use. Ten healthy participants with intact limbs performed five ADLs: drinking from a cup, using a fork, lifting a box, reaching overhead, and hammering, both with and without the prosthesis. Results showed a significant increase in muscle activation when using the prosthesis, particularly in the trapezius (195 %) and deltoid (300 %) muscles, accompanied by increased movement intensity at the shoulder and elbow joints and reduced wrist mobility. These findings indicate fatigue-driven compensatory strategies characterized by proximal overuse and distal restriction. Movement velocity decreased in most activities, with the largest reductions observed during reaching (27 %) and lifting (22 %). RPE values were significantly higher in the prosthesis condition, especially during the fork and reach tasks. Additionally, a reduction in the number of task repetitions was observed when using the prosthesis, particularly during the fork task. Notably, high fatigue levels negatively affected the performance of the sEMG-controlled prosthesis. These results highlight the need for strategies to mitigate MF and improve prosthesis usability during ADLs.

I. INTRODUCTION

Muscle fatigue (MF) is a multifactorial phenomenon influenced by physiological, biomechanical, and task-related factors, including muscle load, repetition, posture, and duration of activity. In upper-limb prosthesis use, additional contributors such as device weight, mass distribution, and control strategies further increase the physical demands placed on the user [1]. Among these, prosthesis weight has

been consistently identified as a primary source of discomfort and fatigue, particularly during prolonged use [2]. Given that prostheses are typically worn for several hours per day, understanding how weight contributes to MF is critical, as fatigue can directly affect motor performance and the reliability of myoelectric control systems [3].

Upper-limb prostheses are designed to restore functionality and independence during Activities of Daily Living (ADLs) [4]. However, despite technological advancements, device abandonment remains a significant issue, with discomfort and fatigue reported as key contributing factors [2], [5]. This highlights the importance of investigating MF under realistic usage conditions to better understand its impact on prosthesis usability and user performance.

Previous studies have explored the relationship between prosthesis use and MF. For instance, *Zheng et al.* investigated the effect of prosthetic weight on MF in upper-limb amputees, showing that increased loading leads to reductions in sEMG frequency features, indicative of fatigue [6]. Similarly, *Díaz-Amador et al.* analyzed the impact of MF on myoelectric control, demonstrating that increasing fatigue significantly reduces classifier accuracy and sensitivity. While these studies provide important insights, they are largely limited to controlled or static conditions [7].

While previous research has demonstrated the negative impact of prosthetic weight on MF and device control, most studies have focused on static conditions. Such approaches fail to capture the dynamic characteristics of ADLs, including continuous joint motion, varying movement velocities, task-dependent load redistribution, and the cumulative effects of repeated movements, all of which are known contributors to MF during real-world prosthesis use. To address this gap, the present study aims to characterize MF induced by prosthetic weight during ADLs. Able-bodied participants were selected for initial testing to better understand the biomechanical and ergonomic demands of prosthesis use. A socket was incorporated to simulate the conditions experienced by individuals with transradial limb loss [8]. The latest version of the PrHand prosthesis [9] was used, weighing 550 g, lighter than the human forearm, which ranges from 755 g to 1,400 g [10].

II. MATERIALS AND METHODS

A. Materials

1) *PrHand Prosthesis*: For this study, the third version of the PrHand prosthesis was used [9]. This device em-

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plays soft robotics and compliant mechanisms. The soft robotic prosthesis features elastic joints that extend the fingers via an internal flexible tendon, while finger flexion is achieved using an inelastic tendon and a servomotor. The fingers are designed with a compliant mechanism that enables Degrees of Freedom (DOF) in flexion/extension, and abduction/adduction. To enhance grasping, the fingers are coated with silicone, increasing friction between the object and the fingers. The prosthesis is controlled via the robot operating system running on a single-board computer (Raspberry Pi 3). An sEMG sensor (Delsys Incorporated, USA) in the digitarium extensor muscle was used to open and close the prosthesis. To allow able-bodied individuals to perform the test, a scaffold was built.

2) *Surface Electromyographic Sensors*: Five sEMG electrodes (Delsys Incorporated, USA) recorded muscle activity from the Trapezius (T), Deltoid (D), Biceps (B), Triceps Long Head (TLH), and Triceps Lateral Head (TLH), selected for their involvement in upper limb movements and compensatory strategies in prosthesis users [6]. The placement of the electrodes followed SENIAM guidelines [11] to ensure reproducibility and minimize cross-talk.

To normalize measurements across participants, a Maximum Voluntary Contraction (MVC) procedure was performed, consisting of three 3-second isometric contractions with 30-second rest intervals, applying standardized resistance to ensure consistency. This procedure was repeated three times for each muscle to account for variability in effort and ensure reliable normalization, following the standardized protocol by Konrad [12]. For sEMG signal acquisition, a sampling frequency of 2000 Hz was used to ensure high-resolution signal capture.

3) *XSens*: The XSens Awinda suit (Movella, Netherlands) was used to capture the movements of a participant during the tests. This MVN Biomechanical motion capture system consists of 10 IMUs positioned on the head, shoulders, upper arms, forearms, hands, sternum and pelvis for building the upper body's biomechanical model. The sensors are wirelessly attached to the body using a shirt and adjustable straps. The placement of the sensor was verified before each trial, this system was chosen to measure compensatory movements [13]. These compensatory strategies can lead to increased MF and joint stress, making it essential to accurately track whole-body kinematics in this study.

B. Subjects

This study involved the voluntary participation of 10 subjects (5 female and 5 male). The inclusion criteria required participants to be healthy individuals aged 18 to 50 years and in a non fatigued state, as determined by the multidimensional fatigue inventory. To establish the non-fatigue status of the subjects, the questionnaire assessed five types of fatigue: general fatigue, physical fatigue, mental fatigue, decreased motivation, and reduced activity.

The exclusion criteria were: subjects with a history of arm fractures, musculoskeletal or systemic disorders, or any known impairment of postural control or motor function. The

TABLE I
DEMOGRAPHIC DATA OF THE SUBJECTS (MEAN $\pm$ STD)

Gender	Age (years)	Weight (kg)	height (cm)	Dominant hand
Female	23.8 $\pm$ 1.6	57.8 $\pm$ 13.0	155.6 $\pm$ 7.8	Right (5)
Male	22.4 $\pm$ 2.4	80.2 $\pm$ 18.6	173.2 $\pm$ 8.3	Right (4) Left (1)

experimental protocol and study objectives were explained to all participants, and informed consent was obtained before the study. The mean and standard deviation (std) of the demographic data for all subjects are presented in Table I. All participants, including the left-dominant individual, used the prosthesis on the right arm to maintain kinematic consistency.

C. Experimental Protocol

This study employed a repeated-measures design to investigate the impact of a prosthetic device on MF during ADLs. Participants performed five activities based on the Activities Measure for Upper Limb Amputation (AM-ULA) [14]. These tasks were specifically chosen to represent a comprehensive range of biomechanical demands: the Cup and Fork tasks (bringing a 200 ml cup to the mouth and mimicking eating with a fork) assessed distal-dominant coordination; the Lift and Hammer tasks (lifting a 1–2 kg box and hammering with a 1 kg weight) evaluated high-load proximal stabilization; and the Reach Overhead task (reaching for a cap at a 2-meter height) tested large-excision movements and compensatory trunk strategies.

Each activity was performed continuously for 2 minutes. No fixed rest periods were provided between the five activities to better simulate the cumulative physical demand of daily life. However, if a participant requested a break or reached a state of significant exertion, a resting period of at least 2 minutes was granted. During the entire test, a maximum of two rest periods was permitted; if a third was required, they were considered completely fatigued and the trial was terminated.

This protocol was adapted from a study that assessed MF in a construction environment by analyzing various work-related activities [15]. Instead of construction tasks, the protocol was tailored to ADLs in a prosthetic service setting. Additionally, previous research has shown that MF can begin to manifest after approximately five minutes of continuous prosthesis use [16], which informed the duration and structure of the experimental design.

Every 30 seconds throughout each 2-minute activity, participants rated their perceived level of MF using the Borg Rating of Perceived Exertion (RPE) scale, ranging from 0 (no exertion) to 10 (maximal exertion), to assess subjective fatigue.

The order of the activities was randomized to minimize order effects and reduce task predictability. Participants always performed all activities first under the no-prosthesis (NP) condition. After resting until full recovery was reported, the same activities were then repeated using the prosthesis (P). Full muscular recovery was ensured by a minimum 10-minute rest period and participant self-confirmation of non-

fatigue before switching conditions. Fig. 1 illustrates the experimental setup, including the placement of the sEMG sensors, the XSens system, and the configuration used to perform the functional tasks.

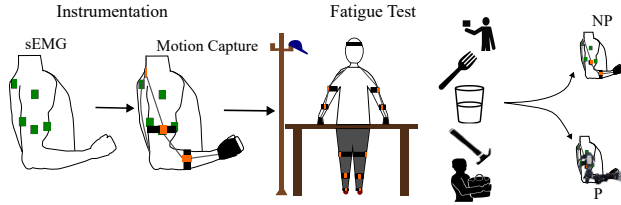


Fig. 1. Experimental protocol for assessing MF during ADLS under NP and P conditions. sEMG and motion capture sensors are used to analyze participant performance during various functional tasks.

The prosthesis is affixed to the user's right arm using a strapping system, consisting of two straps on the forearm and two on the upper arm. These straps are tensioned to ensure a secure and stable fit, minimizing movement. Furthermore, to prevent the prosthesis from slipping off the user's arm, a shoulder strap is applied across the chest, helping to stabilize the device and distribute its weight evenly.

Once the prosthesis is mounted, the user undergoes a muscle signal calibration process. This is done by performing three fist clenches, each held for one second, with ten-second rest intervals between each clench. The resulting sEMG data is then processed to determine the mean muscle activation threshold required for hand opening and closing.

D. Data processing

1) *sEMG Processing*: For sEMG signal processing, the first step was applying a bandpass filter between 15 and 450 Hz to remove unwanted noise. Frequencies below 15 Hz are typically associated with movement artifacts and baseline drift, while those above 450 Hz contain high-frequency noise from external sources. Once the signal is filtered, rectification is performed by taking the absolute value of the signal to ensure all data points are positive [12]. To further smooth the signal and highlight activation patterns, a moving window filter with a window length of 200 milliseconds is applied. Normalization is then performed using the MVC, this allows the comparison between the users. Finally, the Root Mean Square (RMS) is computed per cycle of each ADL to quantify muscle activation [17].

2) *XSens processing*: Joint kinematics were extracted for the right upper limb and analyzed at the shoulder, elbow, and wrist joints. For the shoulder joint, three DOFs were analyzed: Flexion/Extension (F/E), Abduction/Adduction (A/A), and Internal/External rotation (I/E). For the elbow and wrist joints, two DOFs were considered: F/E and Pronation/Supination (P/S). These DOFs were selected as they represent the primary movements involved in the functional tasks. Per repetition of the functional activities, the Range of Motion (ROM), defined as the difference between maximum and minimum joint angles, and the RMS of the joint angle signals were calculated for each DOF. ROM quantified

the angular excursion, while RMS provided a measure of movement intensity and variability during task execution.

In addition, movement velocity was evaluated using the Y-axis angular velocity of the right forearm, as this axis corresponded to the primary direction of motion across tasks. A cycle counter was implemented to track the number of repetitions performed under both conditions (with and without the prosthesis). Movement cycles were identified using a sign-change detection algorithm applied to the velocity signal, where one complete repetition corresponded to four sign changes. Since eccentric contractions generate higher muscle forces than concentric contractions [18], the analysis focused on positive velocity values (upward movements) to enhance sensitivity to fatigue-related changes. To reduce inter-cycle variability, velocity values were averaged for each repetition. The average velocity across the entire session was compared between conditions to assess the impact of prosthesis use on movement efficiency.

3) *Borg Scale*: The Borg values recorded during the test were used to evaluate the variation in perceived exertion while performing activities with and without the prosthesis. The variation in perceived exertion was calculated for each activity by subtracting the initial Borg value from the subsequent ratings for each participant. The perceived exertion was averaged between the participants every 30 seconds.

4) *Statistical analysis*: The null hypothesis states that there is no significant difference in each variable when using the prosthesis compared to not using it. For statistical analysis, data normality was assessed using the Shapiro-Wilk test. To test the initial hypothesis, the choice of statistical test depended on the normality of the data: when the data followed a normal distribution, the Paired Samples t-test was used; otherwise, the Wilcoxon Signed-Rank test was applied. Statistical analysis was performed using R-Studio, with a significance level of 5 % ($p < 0.05$).

III. RESULTS

Fig. 2 presents the five ADLs performed in this study: Fork, Cup, Lift, Reach Overhead, and Hammer. Fig. 3 illustrates an example of the RMS voltage per muscle for one participant during the Cup task, where the increase in muscle activation in P is visible. Table II summarizes the mean normalized RMS values for each activity under NP and P conditions. Percentage Change (PC) relative to NP and p-values from the statistical analysis are also reported, with statistically significant differences highlighted in bold.

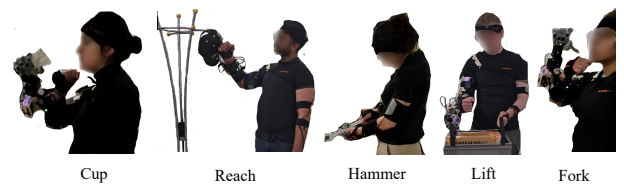


Fig. 2. Five ADLs performed with the prosthesis during the test.

RMS values were generally higher in the P condition, particularly for the T, D, and B muscles across most ADLs

TABLE II

NORMALIZED SEMG RMS VALUES, P-VALUE (P-VAL), AND PERCENTAGE CHANGE (PC) FOR THE TRAPEZIUS (T), DELTOID (D), BICEPS (B), TRICEPS SHORT HEAD (TSH) AND TRICEPS LATERAL HEAD (TLH) MUSCLES DURING FIVE ADLS, UNDER BOTH CONDITIONS NP AND P.

		Cup			Fork			Lift			Reach			Hammer		
		RMS	p-val	PC (%)	RMS	p-val	PC (%)	RMS	p-val	PC (%)	RMS	p-val	PC (%)	RMS	p-val	PC (%)
T	NP	0.10 ± 0.08			0.14 ± 0.10			0.14 ± 0.07			0.19 ± 0.09			0.09 ± 0.04		
	P	0.19 ± 0.10	0.02	93.92	0.23 ± 0.11	0.06	65.32	0.28 ± 0.16	0.05	98.38	0.30 ± 0.12	<0.01	59.74	0.25 ± 0.12	<0.01	191.80
D	NP	0.03 ± 0.01			0.05 ± 0.04			0.08 ± 0.08			0.14 ± 0.07			0.05 ± 0.02		
	P	0.13 ± 0.06	<0.01	301.49	0.11 ± 0.05	0.01	119.52	0.12 ± 0.06	0.07	45.05	0.23 ± 0.12	<0.01	69.90	0.12 ± 0.05	0.01	162.68
B	NP	0.10 ± 0.07			0.19 ± 0.15			0.22 ± 0.23			0.12 ± 0.08			0.13 ± 0.07		
	P	0.28 ± 0.36	0.09	180.94	0.21 ± 0.14	0.63	12.97	0.28 ± 0.26	0.02	26.48	0.19 ± 0.12	0.03	63.16	0.21 ± 0.12	0.01	64.67
TSH	NP	0.07 ± 0.12			0.14 ± 0.30			0.25 ± 0.62			0.08 ± 0.09			0.08 ± 0.15		
	P	0.19 ± 0.39	0.01	163.17	0.20 ± 0.42	0.02	41.64	0.12 ± 0.20	0.14	-52.37	0.22 ± 0.41	0.16	168.72	0.24 ± 0.54	0.01	179.24
TLH	NP	0.21 ± 0.15			0.31 ± 0.18			0.40 ± 0.31			0.25 ± 0.16			0.26 ± 0.13		
	P	0.47 ± 0.38	0.02	121.68	0.50 ± 0.46	0.12	61.02	0.50 ± 0.38	0.04	25.00	0.36 ± 0.21	<0.01	46.13	0.40 ± 0.24	0.02	55.08

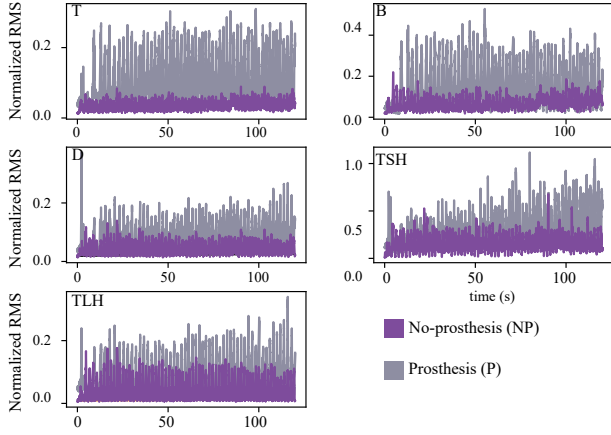


Fig. 3. Muscular activity during Cup task for one user. Purple line represents NP condition, while gray line represents P condition.

(Table II). The largest increases were observed during Hammer for T and Cup for D and B, indicating increased proximal muscular demand during prosthesis-assisted tasks. Significant changes were also observed for TSH and TLH depending on task requirements.

Fig. 4 presents the biomechanical analysis of the shoulder, elbow, and wrist joints during the five ADLs. RMS values and ROM are shown for both NP and P conditions. RMS represents overall joint movement intensity, while ROM reflects total angular excursion. Statistically significant differences between conditions are indicated with an asterisk (*).

At the shoulder joint, RMS values for F/E were higher in P across all activities. Notable increases were observed during the Cup task (from 17.31 to 29.68), Lift (from 17.34 to 28.77), and Hammer (from 11.88 to 20.94). Significant differences were also observed for A/A and I/E in specific tasks, particularly during Reach. ROM analysis showed significant increases in shoulder F/E during the Cup and Fork tasks, while Reach exhibited high ROM values in both conditions.

For the elbow joint, RMS values for F/E were higher in P across all activities, with differences observed during Fork (from 15.23 to 29.63) and Reach (from 37.08 to 40.92). In contrast, RMS values for P/S were lower in the P condition for all tasks. ROM analysis showed task-dependent effects, with significant increases in elbow P/S ROM during Fork and Reach, and reduced F/E ROM during Lift and Reach.

At the wrist joint, RMS values for F/E and P/S were

generally lower in the P condition, particularly during the Reach and Hammer tasks. ROM at the wrist was consistently reduced when using the prosthesis across all activities. Large reductions in F/E ROM were observed during Cup (from 20.87 to 13.61) and Hammer (from 27.77 to 11.76), with statistically significant differences between conditions.

Fig. 5 presents the mean velocity for each activity under NP and P conditions. Purple bars correspond to the NP condition and gray bars to the P condition. Standard deviations are shown for each condition, and the PC relative to NP is displayed for each task. No statistically significant differences in mean velocity were observed between conditions. Mean velocity decreased in the P condition for most activities.

Mean velocity decreased in the P condition for most activities, with the largest reductions observed during Reach (27 %), Lift (22 %), and Hammer (17 %). In contrast, mean velocity increased during the Cup (2 %) and Fork (17 %) tasks. Error bars indicate inter-participant variability.

Fig. 6 presents an example of the variation in RPE based on the Borg scale over time for one of the tasks. The data compares NP (purple dashed line) and P (gray dot line), with mean values and standard deviations represented.

Table III summarizes the rate of change in RPE ($\Delta B/\Delta t$), the PC relative to NP, and the total number of repetitions performed for each task and condition. For all activities, RPE increase was higher in P than in NP. The highest PC values were observed for the Fork and Reach tasks, while smaller differences were observed for Lift and Hammer. It also reports the number of repetitions performed per task. For all activities, fewer repetitions were completed in P than in NP. The largest difference was observed during the Fork task, with 114.38 ± 20.40 repetitions in NP and 87.38 ± 26.53 repetitions in P. The smallest difference was observed during the Reach task, with 74.25 ± 33.88 repetitions in NP and 71.38 ± 24.38 repetitions in P.

IV. DISCUSSION

Muscle activation was consistently higher under the P condition, particularly for the T, D, and B muscles, indicating increased muscular demand during task execution. This behavior is consistent with compensatory strategies previously reported to accommodate the altered mass distribution and inertia associated with upper-limb prostheses [6].

Biomechanical analysis supports these findings, showing increased shoulder and elbow involvement together with

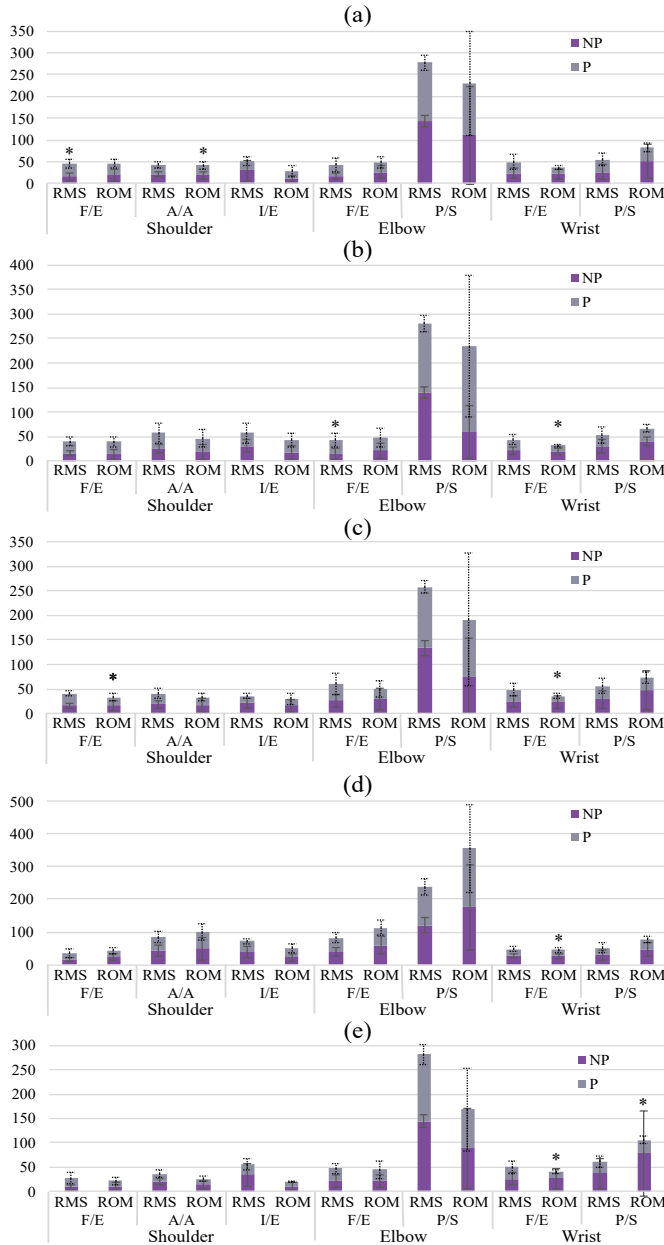


Fig. 4. Biomechanical analysis of shoulder, elbow, and wrist joints during the five ADLs: (a) Cup, (b) Fork, (c) Lift, (d) Reach and (e) Hammer. RMS and ROM are shown for NP and P conditions. Asterisks (*) indicate statistically significant differences between conditions.

reduced wrist mobility, consistent with a proximal redistribution of movement demand during repetitive tasks. The pronounced increase in T activation during Hammer and elevated D and B activity during Cup suggest greater reliance on proximal musculature to compensate for constrained distal motion.

Movement velocity generally decreased under the P condition, particularly during Reach and Lift, which is consistent with increased load and modified joint mechanics reported in prosthesis users [3]. In contrast, Cup and Fork showed slight increases in velocity, likely reflecting altered movement strategies and greater reliance on whole-arm movements

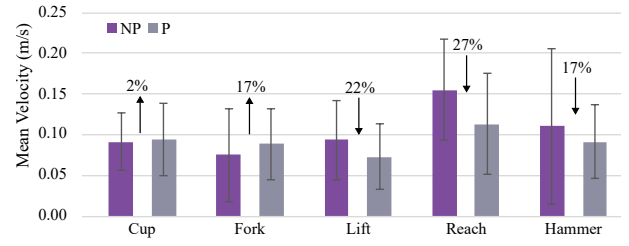


Fig. 5. Mean velocities and standard deviation are presented per activity in P (orange bars) and NP (blue bars) conditions. The Percentage of Change is shown between the conditions per task.

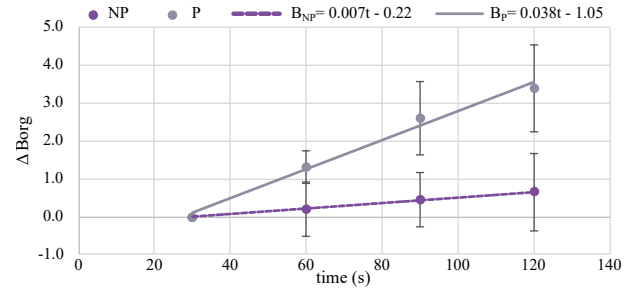


Fig. 6. Example of the variation in RPE based on the Borg scale over time for Reach activity in P (gray line) and NP (purple dashed line) conditions. The values are the average between the participants, the vertical lines are the standard deviation and, the dot lines are the tendency.

rather than forearm-dominant strategies [19], [20]. Task-specific variability in muscle recruitment further suggests that fatigue assessment benefits from multi-muscle sEMG analysis rather than reliance on a single muscle.

Higher rates of RPE increase and fewer repetitions were observed under the P condition across all tasks, indicating earlier fatigue development during prosthesis-assisted activities. The convergence of increased muscle activation, altered joint mechanics, reduced movement efficiency, and elevated RPE provides consistent evidence of fatigue development during repetitive ADLs. Moreover, fatigue may negatively affect sEMG-based prosthesis control performance, increasing unintended activations and reducing reliability [7].

While bypass-socket experiments do not fully replicate clinical biomechanics, they provide a reproducible, controlled framework for isolating prosthesis-specific variables. Prior work confirms that naive users operating a transradial bypass socket achieve functional performance comparable to amputees, validating this configuration as a reliable baseline approximation [8]. Although the 550 g distal mass acts as an additional external load rather than a replacement, introducing joint-axis misalignment and increased effective limb length, the protocol intentionally evaluated the multifactorial interaction of weight, constrained ROM, and myoelectric control effort during realistic ADLs. Isolating mass alone via distributed weights would fail to capture these systemic control and manipulation demands.

Consequently, these findings demonstrate that mitigating fatigue requires a joint mechanical and control optimization approach. Mechanically, reducing distal mass minimizes proximal compensatory torque and delays fatigue onset.

TABLE III

RATE OF CHANGE IN RPE ($\Delta B/\Delta t$) BASED ON THE BORG SCALE IS PRESENTED FOR BOTH CONDITIONS P AND NP, ALSO IT IS SHOWN THE PC RELATIVE TO NP AND THE NUMBER OF REPETITIONS PER TASK.

Tasks		$\Delta B/\Delta t$ (Borg/s)	PC (%)	# Repetitions
Cup	NP	0.000	100.00	86.28 ± 43.21
	P	0.024		66.78 ± 24.63
Fork	NP	0.003	478.13	114.38 ± 20.40
	P	0.019		87.38 ± 26.53
Lift	NP	0.021	44.93	91.63 ± 39.20
	P	0.030		79.50 ± 19.36
Reach	NP	0.007	424.66	74.25 ± 33.88
	P	0.038		71.38 ± 24.38
Hammer	NP	0.012	99.15	99.38 ± 24.74
	P	0.024		85.88 ± 25.00

Simultaneously, implementing adaptive, fatigue-aware myoelectric controllers, shared-control frameworks, or multi-modal sensing configurations can preserve terminal device reliability. In this context, integrating assistive supportive technologies, such as upper-limb weight-bearing exoskeletons, emerges as a highly promising paradigm to actively offload proximal muscle stress and maximize long-term device usability.

V. CONCLUSION

This study evaluated the effects of upper-limb prosthesis use during ADLs by analyzing muscle activation, movement velocity, and RPE. The results establish a baseline in able-bodied participants and highlight key neuromuscular and biomechanical adaptations associated with prosthesis use. Increased activation of the T and D muscles across most tasks indicates substantial shoulder overload, reflecting a shift toward proximal compensation. This was accompanied by reduced distal mobility, decreased movement efficiency in tasks involving large arm excursions, and higher perceived exertion, ultimately leading to faster muscle fatigue development. At higher fatigue levels, sEMG-based prosthesis control deteriorated, resulting in unintended activations and reduced reliability.

These findings emphasize the need to jointly address mechanical and control-related factors in prosthesis design. From a mechanical perspective, improving weight distribution and reducing distal loading may help mitigate compensatory demands. From a control perspective, adaptive and fatigue-aware myoelectric strategies, as well as multi-modal sensing approaches combining sEMG and kinematic data, may enhance robustness during prolonged use. Shared-control strategies may further reduce user effort and improve performance stability.

While this study provides a controlled baseline, validation with individuals with limb loss is essential, as factors such as muscle atrophy, long-term adaptations, and prosthesis experience may influence fatigue development and control performance. Future work should explore different prosthesis designs, including body-powered systems, and investigate assistive solutions such as upper-limb exoskeletons to reduce physical burden and improve long-term usability.

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